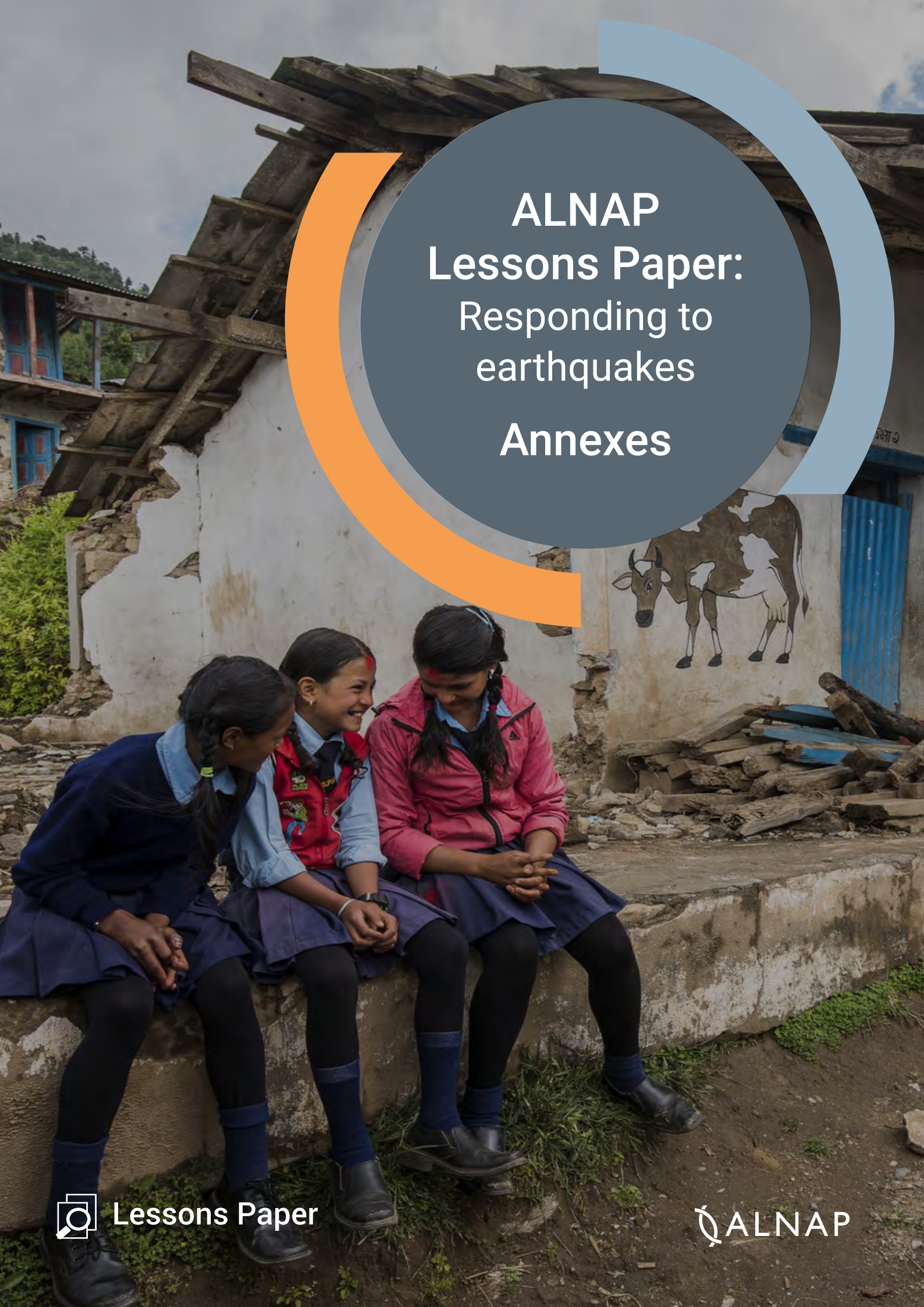




ALNAP Lessons Paper: Responding to earthquakes Annexes



Lessons Paper

ALNAP

Annex 1:

Detailed Methodology

1.1 Inception report

The ALNAP Lessons Paper that these annexes accompany is the result of a structured review of available literature conducted between June and August 2018. The methodology used is aligned with the one described in a recent ALNAP Methods Note (Dillon and Campbell, 2018). A detailed inception report was prepared by Key Aid Consulting and reviewed by ALNAP in June 2018.

1.2 Inclusion criteria

Type of studies

To answer our research questions, we included both qualitative and quantitative research. Because of the overall scarcity of evidence in the humanitarian sector (Puri et al., 2014), we opted for broader inclusion criteria than those traditionally recommended in order to undertake systematic review (Higgins and Green, 2011).

To be eligible for inclusion, qualitative studies had to:

- State they are based on data collected from project stakeholders (affected communities, beneficiaries, implementing agencies, local authorities, etc.).
- Clearly describe the project's inputs, activities, outputs and outcomes.

To be eligible for inclusion, quantitative studies had to apply a research design able to minimise bias in the attribution of identified effects to the applied response – for example, experimental designs (e.g. randomised controlled trials (RCTs) or quasi-experimental designs such as propensity score matching). Mixed-method approaches were also eligible, and those presenting a critical risk of bias have been treated as non-random studies.

We considered including academic articles, literature reviews and grey literature such as evaluations or research papers (published and unpublished) produced by NGOs, international organisations, government agencies and think tanks. Guidance documents have been included only if they make it clear they have been developed based on documented lessons learnt.

Publication types considered ineligible include: personal blogs, commentaries, diaries, opinion pieces, workshop reports, marketing material such as 'life stories', newspaper articles, magazine articles, and legal proceedings/court documents.

We included studies from 2008 onwards, as ALNAP's 2008 earthquake paper already captured the lessons from papers published before that date.

Searches were conducted in English, but studies were not excluded on the basis of language, as documents were still considered if a French or English translation existed.

Type of populations and contexts

This lessons paper focuses on responses that attempt to provide support to crisis-affected households in all locations affected by earthquakes since 2008. The populations of interest are those that have been affected directly and indirectly by any earthquake since 2008 as long as the earthquake was of such a scale that it can be considered a humanitarian crisis.

All crisis-affected population groups have been included irrespective of their age, gender, disability, health status, literacy, pregnancy status, parental status, specific status according to the Geneva Conventions (and their additional protocols), ethnic or religious belonging, and level of income. Location has not been used as an exclusion criterion. Studies not reporting on a specific earthquake or range of earthquakes have been excluded.

Type of response

This lessons paper focuses on sectoral and multi-sectoral earthquake responses. Studies were included if the described response(s) aimed to: (a) improve basic needs coverage, (b) restore/rebuild livelihoods, or (c) reduce negative coping mechanisms. We have not used a restrictive definition of negative coping mechanisms and included all responses that state they intend to reduce negative coping mechanisms. If a response contains several components, including at least one of the above, we have considered it for inclusion. Studies exclusively covering earthquake preparedness have not been included for the sake of a homogenous final analysis.

Studies describing responses delivering any type of activities have been included as long as the activities aimed to improve the situation of earthquake-affected populations.

We have included literature about post-earthquake responses designed by any actors (NGOs, UN agencies, Red Cross/Red Crescent movement, private sector actors and government actors) in the aftermath of an earthquake in order to benefit crisis-affected people. Studies were not excluded on the basis of the response length.

Type of outcomes

Documents that identify any type of outcomes measures have been included. We also included studies not measuring outcomes but measuring process instead.

We did not place any specific restriction on the way the outcome was measured or the duration of the outcome measurement. Therefore, we did not exclude any study from the lessons paper on the basis of unreliable measurement outcomes.

1.3 Search method

To be structured and comprehensive, our search strategy included academic databases and journals as well as institutional websites. The search was undertaken between July and August 2018.

We searched all the databases included in [Section 2.4 Database search](#). For each database, we used search strings based on a declination of four concepts:

- earthquake
- humanitarian response
- sectoral/multi-sectoral
- lessons.

We used Boolean operator 'OR' to link each key aspect to their synonyms, and operator 'AND' to combine several notions. (See Section 2.1 for the key words and examples of search strings, and Section 2.4 for all database searches conducted.)

Most of the libraries of the websites we searched manually use simple search functions. Therefore, we adjusted our search strings and for some of them used a single search term – 'earthquake' – to ensure that we comprehensively screened all existing literature.

Beyond electronic searches, we also reached out to a number of sources that may provide relevant literature on earthquake responses:

- We interviewed three people (David Sanderson, Abhijit Bhattacharjee and Maggie Stephenson) and asked for any relevant studies.
- We reached out to practitioners and asked for relevant studies via a message posted on Pelican and Urban response D-groups. We also posted a message on the evaluation D-Group to specifically ask for lessons on the evaluation of earthquake responses.
- We conducted a forward citation search via Google Scholar to search for all studies citing the 65 included studies.
- A backward citation search was conducted: the research team reviewed the bibliography of each document rated as fit for inclusion to find further studies that the designed search strings may not have detected. No additional studies were found.

1.4 Data collection and analysis

Selection of studies

The screening and selection process took place in two stages. Exclusion filters are detailed in the inception report.

As a first step, the research analyst screened all titles and abstracts of the documents resulting from the search. Studies were classified as either 'excluded' or 'included for step 2'. All studies classified as 'included for step 2' were registered in Zotero. They were then screened by the same researcher. Studies were again classified as either 'included' or 'excluded', with the reason for exclusion recorded. See Section 2.5 for a list of all excluded studies.

Data extraction

The research team developed a detailed coding sheet (see Section 2.6). The research analyst extracted and coded all the included studies. The information extracted included information about the population, the response, the context, the outcomes and the lessons learnt.

Quality appraisal

After data extraction, we appraised the quality of each included study. To reduce the risk of bias, 5% of the studies were appraised by both researchers to compare results and ensure consistency in ranking. The level of agreement was high considering the clear appraisal criteria set in the tool.

For the purpose of this evidence synthesis, a specific quality appraisal template was developed (see Section 2.7). It is a mixed-methods critical appraisal tool with three modules: one each for qualitative studies, quantitative studies and mixed-methods studies.

Data synthesis

An overview of the data extracted was prepared and organised as per the different pre-identified steps of the project cycle. It served to identify trends and prepare for the thematic synthesis. Studies identified to address the research questions are heterogeneous in terms of design but consistent in terms of lessons learnt, and we found few contradictions across the included studies. When relevant, data has been analysed by gender, age and ethnicity. As such, it has been possible to synthesise lessons based on a satisfying number of studies. See Section 2.8 for a detailed overview of which paper contributed to which lesson.

Finally, we compared the lessons from the 2008 paper and those originating from this search. Despite differences in scope, findings are rather aligned between both papers and lessons included are to some extent similar. See Section 2.3 for a detailed comparison.

1.5 Search results

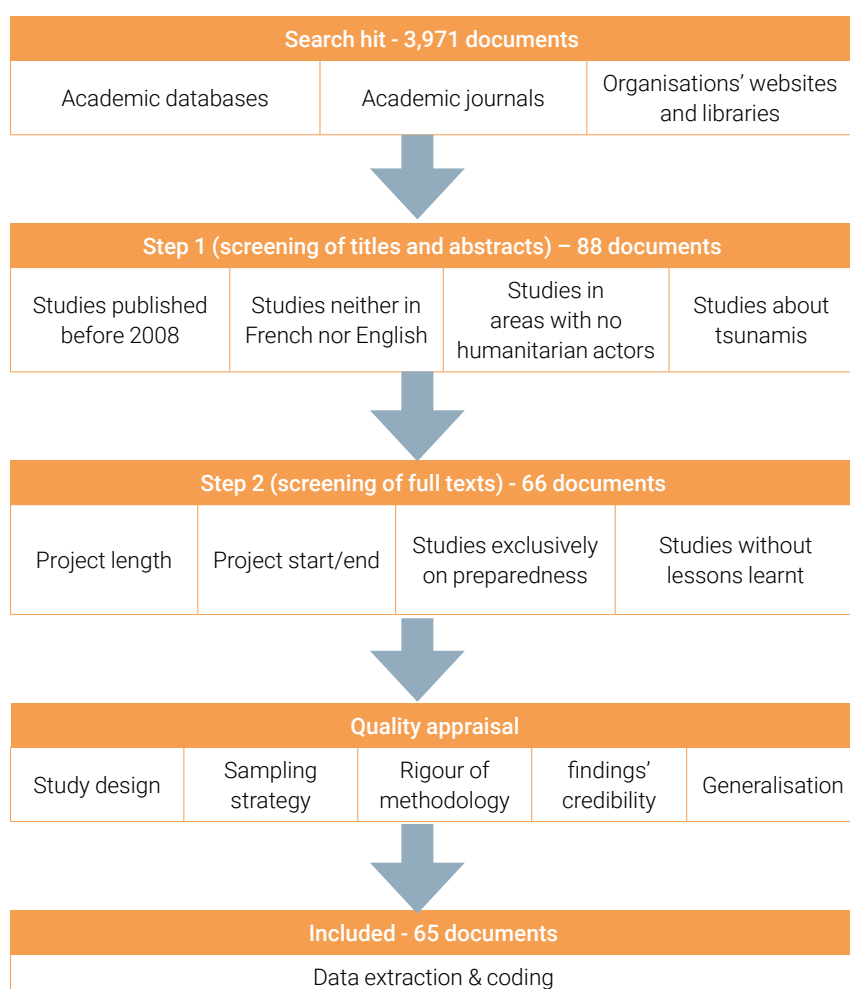
Key steps of the search

We identified 3,971 records but excluded 3,832 after title and abstract screening as they were irrelevant to the research question. All documents were then included in Zotero, which identified 51 duplicates. We then obtained full texts of the remaining 88 studies. As part of the second step, we excluded a further 22 studies, mostly because they either featured no lessons learnt (9) or because they were outside of the scope (7), or both (2). Other reasons for exclusion were:

- document contains terms of reference only
- document is only constituted of minutes from a meeting
- document is a blog/an opinion piece
- document is a PowerPoint presentation
- lessons included are too specific to a particular body, not applicable to humanitarian practitioners.

As a result, 66 documents went for data extraction and underwent a quality appraisal. Finally, 65 documents were selected for inclusion. Figure 1 summarises the key steps of the search.

Figure 1: Key steps of the search strategy



Characteristics of included papers

We included 65 studies reporting on earthquakes responses: 5 are identified as mixed methods, 60 as qualitative and none as quantitative. The documents included contain 23 evaluation papers, 15 journal articles, 11 working papers, 7 lessons papers, 7 research reports, 1 advocacy paper and 1 thesis (see Table 1).

Table 1: Type of study included in the lesson paper

Document and author	Type of study	Quality appraisal score (out of 48)
Altay, N. and Labonte, M. (2014) 'Challenges in humanitarian information management and exchange: evidence from Haiti'. <i>Disasters</i> , 38(s1): s50-S2 (https://doi.org/10.1111/disa.12052)	Journal article	32
Amnesty International (2011) 'Aftershocks: women speak out against sexual violence in Haiti's camps' (https://resourcecentre.savethechildren.net/node/3999/pdf/3999.pdf)	Advocacy paper	12
Audefroy, J.F. (2011) 'Haiti: post-earthquake lessons learned from traditional construction'. <i>Environment and Urbanization</i> , 23(2): 447-462 (https://doi.org/10.117)	Journal article	20
Baker, J. and Narayanan, U. (n.d.) <i>Independent evaluation of the NRC expert deployment/NORCAP response to the Nepal 2015 earthquake</i> (https://www.nrc.no/globalassets/pdf/evaluations/norcap-nepal-evaluation-report_final.pdf)	Evaluation report	22
Barber, R. (2016) <i>Did the humanitarian response to the Nepal earthquake ensure no one was left behind? A case study on the experience of marginalised groups in humanitarian action</i> . Save the Children (https://resourcecentre.savethechildren.net/node/9901/pdf/nepal_earthquake_gesi_report_march_2016.pdf)	Evaluation report	24
Bhattacharjee, A. (2016) <i>After-action review of Nepal earthquake response</i> . UNDP Crisis Response Unit (https://www.alnap.org/help-library/after-action-review-of-nepal-earthquake-response)	Evaluation report	33
Bhattacharjee, A. and Lossio, R. (2011) <i>Evaluation of OCHA response to the Haiti earthquake. Final report</i> (https://www.unocha.org/sites/dms/Documents/Evaluation%20of%20OCHA%20Response%20to%20the%20Haiti%20Earthquake.pdf)	Evaluation report	44
Bisri, M. and Beniya, S. (2016) 'Analyzing the national disaster response framework and inter-organizational network of the 2015 Nepal/Gorkha earthquake'. <i>Procedia Engineering</i> , 159: 19-26 (https://doi.org/10.1016/j.proeng.2016.08.059)	Journal article	27
Brady, C. (2012) <i>An Emergency Market Mapping and Analysis study: changing responses to the Haiti earthquake</i> . Oxford: Oxfam GB (https://logcluster.org/sites/default/files/emma_haiti_case_study_2014.pdf)	Evaluation report	24
Chaudhary, P., Vallese, G., Thapa, M., Alvarez, V.B., Pradhan, L.M., Bajracharya, K., Sekine, K., Adhikari, S., Samuel, R. and Goyet, S. (2017) 'Humanitarian response to reproductive and sexual health needs in a disaster: the Nepal earthquake 2015 case study'. <i>Reproductive Health Matters</i> , 25(51): 25-39 (https://doi.org/10.1080/09688080.2017.1405664)	Journal article	33

Chen, K., with Zhang, Q. and Hsu, C. (2016) <i>Earthquake lessons from China: coping and rebuilding strategies</i> . Washington DC: International Food Policy Research Institute (IFPRI)	Research report	39
Christian Aid (2012) <i>Haiti: unconditional cash transfers – lessons learnt</i> . London: Christian Aid (https://www.christianaid.org.uk/sites/default/files/2016-03/haiti-unconditional-cash-transfers-briefing-paper-jan-2012.pdf)	Evaluation report	10
de Ville de Goyet, C., with Sarmiento, J. and Grünewald, F. (2011) <i>Health response to the earthquake in Haiti, January 2010 - lessons to be learned for the next massive sudden-onset disaster</i> . Pan American Health Organization (http://new.paho.org/disasters/dmdocuments/HealthResponseHaitiEarthq.pdf)	Evaluation report	20
Clermont, C., with Sanderson, D., Sharma, A. and Spraos, H. (2011) <i>Urban disasters – lessons from Haiti: study of member agencies' responses to the earthquake in Port Au Prince, Haiti, January 2010</i> (https://www.alnap.org/help-library/urban-disasters-%E2%80%93-lessons-from-haiti-study-of-member-agencies%E2%80%99-responses-to-the)	Research report	24
Cohen, M. (2012) <i>Haiti: the slow road to reconstruction – two years after the earthquake</i> . Oxfam Briefing Note. Oxford: Oxfam International (https://d1tn3vj7xz9fdh.cloudfront.net/s3fs-public/file_attachments/haiti-reconstruction-two-years-after-earthquake-100112-en_4.pdf)	Evaluation report	4
Cohen, M. and Figaro, J. (2014) <i>Urban disaster response and recovery: gender-sensitive WASH programming in post-earthquake Haiti</i> . Oxford: Oxfam GB (https://oxfamlibrary.openrepository.com/oxfam/handle/10546/311058)	Evaluation report	4
Cook, A., with Shrestha, M. and Htet, Z. (2018) 'An assessment of international emergency disaster response to the 2015 Nepal earthquakes'. <i>International Journal of Disaster Risk Reduction</i> , 31: 535-547 (https://doi.org/10.1016/j.ijdr.2018.05.014)	Journal article	31
Cosgrave, J. (2008) <i>Responding to earthquakes 2008. Learning from earthquake relief and recovery operations</i> . London: ALNAP/ODI (https://www.alnap.org/help-library/responding-to-earthquakes-learning-from-earthquake-relief-and-recovery-operation)	Lessons paper	10
Davidson, S. (2011) <i>A review of the IFRC-led shelter cluster</i> . IFRC (https://www.alnap.org/help-library/a-review-of-the-ifrc-led-shelter-cluster-haiti-2010)	Evaluation report	30
Deters, L. (2011) <i>The use and impact of ECD kits: post-earthquake Haiti 2010</i> . UNICEF (https://www.unicef.org/evaldatabase/index_66807.html)	Evaluation report	35
Dolan, C. and Ververs, M. (n.d.) 'The Haiti earthquake – country and global level cluster coordination experiences and lessons learnt' (https://www.enonline.net/fex/39/haiti)	Evaluation report	4
Elhawary, S. and Castillo, G. (2008) <i>The role of the affected state: a case study on the Peruvian earthquake response</i> . HPG Working Paper. London: Humanitarian Policy Group, Overseas Development Institute (ODI) (https://www.odi.org/sites/odi.org.uk/files/odi-assets/publications-opinion-files/1838.pdf)	Working paper	18
Gerdin, M., Wladis, A. and von Schreeb, J. (2012) 'Foreign field hospitals after the 2010 Haiti earthquake: How good were we?' <i>Emergency Medicine Journal</i> , 30(1): e8 (https://doi.org/10.1136/emmermed-2011-20071)	Journal article	40

Grünewald, F. and Burlat, A. (n.d.) 'Groupe URD - tremblement de terre au Népal: une synthèse de la réponse et quelques leçons apprises' (https://www.urd.org/Tremblement-de-terre-au-Nepal-une)	Working paper	5
Grünewald, F., Kauffmann, D., Boyer B. and Patinet, J. (2011) <i>Real-time evaluation of humanitarian action supported by DG ECHO in Haiti : 2009-2011</i> . Paris: Groupe URD (https://www.alnap.org/help-library/real-time-evaluation-of-humanitarian-action-supported-by-dg-echo-in-haiti-2009-2011)	Evaluation report	34
Guha-Sapir, D., Kirsch, T., Dooling, S. and Sirois, A. (2011) <i>Independent review of the U.S. government response to the Haiti earthquake</i> . United States Agency for International Development (USAID)/Office of Foreign Disaster Assistance (OFDA) (https://pdf.usaid.gov/pdf_docs/pdacr222.pdf)	Evaluation report	30
Hall, M.L., Lee, A.C.K., Cartwright, C., Marahatta, S., Karki, J. and Simkhada, P. (2017) 'The 2015 Nepal earthquake disaster: lessons learned one year on'. <i>Public Health</i> , 145: 39-44 (https://doi.org/10.1016/j.puhe.2016.12.031)	Journal article	27
Hartberg, M., Proust, A. and Bailey, M. (2011) <i>From relief to recovery: supporting good governance in post-earthquake Haiti</i> . Oxford: Oxfam (https://policy-practice.oxfam.org.uk/publications/from-relief-to-recovery-supporting-good-governance-in-post-earthquake-haiti-118929)	Working paper	7
Hirano, S. (2012) <i>Learning from the urban transitional shelter response in Haiti</i> . Catholic Relief Services (https://www.crs.org/our-work-overseas/research-publications/learning-urban-transitional-shelter-response-haiti)	Evaluation report	26
Inter-Agency Standing Committee (2010) <i>Response to the humanitarian crisis in Haiti: achievements, challenges and lessons to be learned</i> . IASC (https://reliefweb.int/report/haiti/response-humanitarian-crisis-haiti-following-12-january-2010-earthquake-achievements)	Evaluation report	28
Jackson, R. (2015) <i>Rebuilding a more resilient Nepal: key recommendations for reconstruction and recovery</i> . Oxford: Oxfam GB (https://policy-practice.oxfam.org.uk/publications/rebuilding-a-more-resilient-nepal-key-recommendations-for-reconstruction-and-re-581286)	Working paper	8
Jackson, R., Fitzpatrick, D., and Man Singh, P. (2016) <i>Building back right: ensuring equality in land rights and reconstruction in Nepal</i> . Oxford: Oxfam GB (https://policy-practice.oxfam.org.uk/publications/building-back-right-ensuring-equality-in-land-rights-and-reconstruction-in-nepal-606028)	Working paper	25
Jha, A. K., Barenstein, J., Phelps, P., Pittet, D. and Sena, S. (2010) <i>Safer homes, stronger communities: a handbook for reconstructing after natural disasters</i> . Washington DC: World Bank (https://openknowledge.worldbank.org/handle/10986/2409)	Lessons paper/manual	23
Kauffmann, D. (2012) <i>Review of cash-transfer coordination in Haiti following the earthquake of January 2010</i> . Cash Learning Partnership (CaLP) (http://www.cashlearning.org/downloads/GroupURD%20-%20Cash%20coordination%20-%20Haiti.pdf)	Evaluation report	34
Kilby, P. and Williamson, K. (2011) 'NGO and community-oriented disaster response: lessons from the humanitarian response to the Asian tsunami in India and the Jogjakarta earthquake in Indonesia' in Shondell Miller, D. and Rivera, J.D. (eds) <i>Comparative emergency management: examining global and regional responses to disasters</i> . CRC Press LLC (https://openresearch-repository.anu.edu.au/handle/1885/17414)	Lessons paper	20

Kirk, J. (2008) <i>Building back better: post-earthquake responses and educational challenges in Pakistan</i> . UNESCO/International Institute for Educational Planning (IIEP)	Research report	29
Klenk, J. (2017) <i>Report on three capitalization workshops focused on good practices and lessons learned by the food security sector</i> (http://fscluster.org/sites/default/files/documents/haiti-final_report_-_capitalisation_workshops-haiti_-_abridged_english_version_0.pdf)	Working paper	23
Lu, Y. and Xu, J. (2015) 'NGO collaboration in community post-disaster reconstruction: field research following the 2008 Wenchuan earthquake in China'. <i>Disasters</i> , 39(2): 258-278	Journal article	34
McGearty, S., O'Hagan, P. and Montinard, M. (2012) "Together we are stronger": an independent final evaluation of the action by Churches Together Alliance Haiti appeal HTI-101. ACT (Action by Churches Together) Alliance (https://www.alnap.org/help-library/together-we-are-stronger-an-independent-final-evaluation-of-the-action-by-churches)	Evaluation report	35
Mughal, H., Ahmed, S.A., Mumtaz, H., Tanwir, B., Bilal, S. and Stephenson, M. (2015) <i>Kashmir earthquake 2005: learning from the shelter response and rural housing recovery</i> (https://www.sheltercluster.org/sites/default/files/docs/shelter_in_recovery_kashmir_eq_2005.pdf)	Lessons paper	7
Nabi, P. (2014) 'Coordinating post-disaster humanitarian response: lessons from the 2005 Kashmir earthquake, India'. <i>Development in Practice</i> , 24(8): 975-988 (https://doi.org/10.1080/09614524.2014.964187)	Journal article	39
Nekoei-Moghadam, M., Amiresmaili, M. and Aradoei, Z. (2016) 'Investigation of obstacles against effective crisis management in earthquake'. <i>Journal of Acute Disease</i> , 5(2): 91-95 (https://doi.org/10.1016/j.joad.2015.10.001)	Journal article	30
Moody, S., Morley, L. and Davidson, C. (2018) Evaluation of the Canterbury earthquake appeal & recovery programme. New Zealand Red Cross (https://media.ifrc.org/ifrc/wp-content/uploads/sites/5/2018/02/CEARP-Technical-Evaluation-CORRECT.pdf)	Evaluation report	39
Oxfam (2016) "I am alone": single women and the Nepal earthquake. Nepal: Oxfam and Women for Human Rights (https://d1tn3vj7xz9fdh.cloudfront.net/s3fs-public/ib-i-am-alone-nepal-earthquake-single-women-230716-en.pdf)	Research report	25
Ansari, A. (2010a) Haiti: a once-in-a-century chance for change - beyond reconstruction: re-envisioning Haiti with equity, fairness, and opportunity. Oxford: Oxfam International (https://www.oxfam.org/sites/www.oxfam.org/files/bp136-haiti-once-in-a-century-220310-summary.pdf)	Working paper	5
Ansari, A. (2010b) Reconstructing Haiti. Oxford: Oxfam International (https://www.oxfam.org/sites/www.oxfam.org/files/reconstructing-haiti-jan-2010.pdf)	Research paper	4
Shui, Q. (n.d.) 'Sichuan earthquake one year report May 2009'	Working paper	4

Rees-Gildea, P. and Moles, O. (2012) <i>The International Federation of Red Cross and Red Crescent Societies shelter programme in Haiti 2010-2012. Lessons learned</i> . International Federation of Red Cross & Red Crescent Societies	Evaluation report	33
Roussel, P. (2016) 'Exploring emergency communication experiences in response to the 2010 Haiti earthquake: lessons learned from international donors and local government agencies'. Theses & Dissertations, May (http://digitalcommons.unmc.edu/etd/87)	Theses and dissertations	42
Sanderson, D. and Ramalingam, B. (2015) <i>Nepal earthquake response: lessons for operational agencies</i> . ALNAP Lessons Paper. London: ALNAP/ODI (https://www.alnap.org/help-library/nepal-earthquake-response-lessons-for-operational-agencies)	Lessons paper	15
Sanderson, D., Knox Clarke, P. and Campbell, L. (2012) <i>Responding to urban disasters: learning from previous relief and recovery operations</i> . London: ALNAP/ODI (https://www.alnap.org/system/files/content/resource/files/main/alnap-provention-lessons-urban.pdf)	Lessons paper	15
Sanderson, D., Sharma, A., Kennedy, J. and Burnell, J. (2014) 'Principles, practice and lessons from Haiti for urban post-disaster shelter recovery programs'. <i>Asian Journal of Environment and Disaster Management (AJEDM)</i> , 6(2): 131-151 (https://doi.org/10.3850/S1793924014000362)	Journal article	15
Save the Children (2015) <i>Haiti earthquake: five years on Results and lessons learned</i> . Save the Children (https://resourcecentre.savethechildren.net/node/8767/pdf/haiti-earthquake-five-years-on.pdf)	Research report	4
Solidarités International (2017) <i>Community approach and institutional support in urban areas: – the experience of Solidarités International in Christ Roi neighbourhood in Port-Au-Prince, Haiti</i> (https://www.solidarites.org/wp-content/uploads/2018/01/Community-approach-and-institutional-support-in-urban-areas-Christ-Roi-Port-au-Prince---Haiti.pdf)	Evaluation report	19
Standing, K., Parker, S. and Bista, S. (2016) 'Grassroots responses to violence against women and girls in post-earthquake Nepal: lessons from the field'. <i>Gender & Development</i> , 24(2): 187-204 (https://doi.org/10.1080/13552074.2016.1194562)	Journal article	20
Tafti, M.T. and Tomlinson, R. (2015) 'Best practice post-disaster housing and livelihood recovery interventions: winners and losers'. <i>International Development Planning Review</i> , 37(2): 165-185 (https://doi.org/10.3828/idpr.2015.14)	Journal article	21
Groupe URD (2011) 'Beyond emergency relief in Haiti - January 2011' (https://reliefweb.int/sites/reliefweb.int/files/resources/E8A7F0BAF073FACC852578260059134B-Full_Report.pdf)	Working paper	20
Vahanvati, M. and Mulligan, M. (2017) 'A new model for effective post-disaster housing reconstruction: lessons from Gujarat and Bihar in India'. <i>International Journal of Project Management</i> , 35: 802-817	Journal article	32

Wendelbo, M., La China, F., Dekeyser, H., Taccetti, L., Mori, S., Aggarwal, V., Alam, O., Savoldi, A. and Zielonka, R. (2016) <i>The crisis response to the Nepal earthquake: lessons learned</i> . European Institute for Asian Studies (http://www.eias.org/wp-content/uploads/2016/02/The-Crisis-Response-to-the-Nepal-Earthquake--Lessons-Learned-colour-1.pdf)	Research report	10
Withers, L. and Dahal, N. (2015) <i>After the earthquake: Nepal's children speak out</i> . Save the Children/Plan International/UNICEF/World Vision International (https://resourcecentre.savethechildren.net/node/9356/pdf/nepal_children_consultation_final_lowres.pdf)	Research report	38
World Bank (2016) <i>What did we learn? The shelter response and housing recovery in the first two years after the 2010 Haiti earthquake</i> . Washington DC: World Bank (http://documents.worldbank.org/curated/en/699481494923590147/pdf/115025-WP-PUBLIC-haiti-what-did-we-learn.pdf)	Lessons paper	23
World Bank (2014) 'How to develop effective disaster relief programs – lessons from vulnerable countries' 10 September (http://www.worldbank.org/en/news/feature/2014/09/10/lessons-in-effective-disaster-recovery-from-vulnerable-countries)	Working paper	2
Young, P. and Henderson, E. (2010) <i>The Haiti earthquake: an urban solution</i> . Oxfam (http://www.cashlearning.org/downloads/resources/casestudies/haiti.pdf)	Working paper	4
Zissman, M., with Evans, J., Holcomb, K. and Jones, D., et al. (2014) 'Development and use of a comprehensive humanitarian assessment tool in post-earthquake Haiti'. <i>Procedia Engineering</i> , 78: 10-21 (https://www.sciencedirect.com/science/article/pii/S1877705814010200)	Journal article	38

Overall, the quality of design and methodology of the included studies was poor. Apart from a few academic studies and some evaluations, most of the documents included little or limited methodology. The main reason for this was that most documents aimed to be use-oriented and provide practical recommendations and lessons to help donors, NGOs, UN agencies, local governments and others involved in earthquake response to improve what they do. Despite this, we decided to keep all documents (but one) after quality appraisal, since all of them presented supporting evidence of some kind (based on experience, interviews, references to past projects) for the lessons contained in them.

We acknowledge the risk of bias in this paper (i.e. underreporting of studies establishing negative or mixed evaluation findings). Nonetheless, this risk appears to be reasonable considering that many (if not most) of the lessons included in this paper are based on failures of the humanitarian community during an earthquake response.

Another potential bias is due to the fact that the results of the database search barely included documents on government-led responses to earthquake. As a result, the paper potentially lacks lessons on government-led reconstruction.

Characteristics of included responses

The research team initially expected to focus on major earthquakes since 2008 (see Table 2).

Table 2: Major earthquakes (2008-2018)

Date	Country	Country GDP per capita (\$/year)	Number of deaths	Magnitude
2018	Papua New Guinea	2192	145	7,5
2017	Iraq/Iran	4000	400	7,3
2017	Mexico	10153	369	7,1
2016	Ecuador	7700	650	7,8
2015	Afghanistan	800	400	7,8
2015	Nepal	1000	9000	7,8
2014	China	6100	600	6,3
2013	Pakistan	2600	825	7,7
2012	Iran	13053	300	6,4
2011	Turkey	12900	600	7,2
2011	Japan	35300	15690	9
2011	New Zealand	36648	180	6,3
2010	Chile	23194	500	8,8
2010	Haiti	1783	100000	7
2008	China	6100	87600	7,8

The lessons included in this paper touch upon the following earthquakes: 2003 Bam Earthquake in Iran, the 2008 earthquake in Peru, the 2008 Wenchuan Earthquake in China, the 2010 earthquake in Haiti, the 2011 earthquake in New Zealand, the 2011 earthquake in Japan, and the 2015 earthquake in Nepal.

As expected, many of the lessons included in this paper focus on evaluations of the response to three of the major (and deadliest) earthquakes during this period: China (2008), Haiti (2010) and Nepal (2015).

Annex 2

Additional information on searches and tools

2.1 Key words and search strings

The authors developed key search terms to address the research question (Table 3).

Table 3: List of key words for search

Earthquake	Humanitarian response	Sectoral/Multi-sectoral	Lessons
tremor Nepal Haiti China	humanitarian relief recovery	food security education livelihood WASH protection nutrition health Multi-sectoral debris management shelter	lessons learnt impact assessment review result consequence evaluation

Based on these key words, the authors developed search strings for each database, whether academic or institutional. The authors used Boolean operator 'OR' to link each key aspect to their synonyms, and operator 'AND' to combine several notions.

For example: (earthquake) AND (humanitarian OR humanitarian response OR relief OR recovery) AND (sectoral OR food security OR livelihood OR WASH OR education OR protection OR shelter OR nutrition OR health OR multi-sectoral) AND (lessons OR lessons learnt OR impact OR assessment OR review OR evaluation).

Search strings were tested and adapted to the specificities of each database. For instance, institutional databases often did not take Boolean operators into account, so we used a simplified search string with just three to four words.

2.2 Quality appraisal tool

The quality of each included document was then assessed using a critical appraisal tool (see Section 2.7), which was based on Langer et al., 2014. It used five criteria:

- study design
- sampling strategy and/or risk of bias
- rigour of the methodology/study process
- findings' credibility
- contextualisation and generalisation of the study findings.

The reviewers measured the achievement and attributed a score per indicator. A score of 0 was attributed if the indicator was not met, a score of 2 if the indicator was met, and a score of 1 if the indicator was partly met. At the end of the process, each of the documents shortlisted had a quality score.

To minimise the risk of personal bias, both reviewers independently implemented the critical appraisal tool for 5% of the studies included (four studies each). They then shared, discussed and agreed the assessment of each document against each of the criteria in order to determine an overall quality score. Despite a few minor differences, both reviewers came up with similar results, which means that the criteria used were sufficiently clear and non-ambiguous for objective quality appraisal.

Although several studies included only very limited detail on study design, the decision was taken to retain all but one of the studies. In particular, most of the lessons papers included barely had a methodology section or discussed any element of study design. Nonetheless, considering that they displayed interesting and credible lessons learnt based on a wide variety of sources, we decided to include them, despite their low score on the quality appraisal tool.

2.3 Consistency/discrepancy with 2008 earthquake lessons paper

This paper is a revision of ALNAP's 2008 lessons paper, *Responding to earthquakes: learning from earthquake relief and recovery operations* (Cosgrave, 2008). The findings of both papers are closely aligned, although we should mention a few differences in scope, methodology, findings and structure.

The most obvious difference is in scope: Cosgrave's paper covered three decades, while this paper covers one (the most recent). Unsurprisingly, this also means that Cosgrave's paper has more examples of earthquakes to draw on. (Cosgrave also included tsunamis, whereas this paper does not.)

There are also significant differences in the methodology used. Cosgrave's paper is a 'thorough rewriting of two previous briefing papers' (one on relief, the other on recovery). The present paper is a systematic review in itself, based on ALNAP's 2018 Method Note for lessons papers.

Both papers are also structured differently. Cosgrave's paper is based on type of response (i.e. recovery vs relief, sectoral as well as a small section on management), whereas this paper is structured around the phases of the project cycle.

Both papers focus on similar sectors – shelter, health, livelihoods, debris management, food security, cash-based assistance – and discuss cross-cutting issues too (protection, gender, and the environment). Many of the lessons highlighted by both papers are similar (e.g. only relocate/resettle as a last resort, consult affected people about their priorities, focus on flexibility/adaptive management). However, based on new evidence available, the present report has some different findings. For instance, the outbreak of disease after an earthquake was considered unlikely in 2008, but the cholera epidemic in Haiti has proved that epidemics can happen in post-earthquake settings.

2.4 Database search

Database name	URLs	Search string used (use one line per search string)	Date & time of the search	Total number of hits	Number of excluded hits - Step 1	Main reasons for exclusion after Step 1	In Zotero
Science Direct	www.sciencedirect.com	earthquake + humanitarian + impact + review + lessons	18/06 @ 16:35	311	305	not relevant	6
ELDIS	www.eldis.org	earthquake AND humanitarian	18/06 @ 17:45	58	56	most documents are prior to 2008	2
ELLA	http://ella.practicalaction.org/	all key words with Boolean operators	29/06 @ 18:47	500	483	not relevant	17
SCOPUS	https://www.scopus.com/	earthquake AND response AND humanitarian AND lessons	29/06 @ 17:30	223	215	outside of scope	8
Bielefeld Academic Search Engine (BASE)	www.base-search.net/Search	earthquake and lessons and humanitarian	29/06 @ 17:52	6	2	not relevant	4
Google Scholar	https://scholar.google.fr/	all key words with Boolean operators	27/06 @ 14:03	500	457	not relevant	43
Development and Change	https://onlinelibrary.wiley.com/doi/10.14677660	earthquake and humanitarian and lessons	29/06 @ 17:42	4	4	not relevant	0
Third World Quarterly	https://www.tandfonline.com/toc/ctwq20/current	earthquake and humanitarian response and lessons	29/06 @ 15:39	22	19	not relevant	3
Disasters	https://onlinelibrary.wiley.com/journal/14677717	earthquake and humanitarian response and lessons	29/06 @ 15:43	1	0	not relevant	1
Development in Practice	https://www.tandfonline.com/loi/cdip20	earthquake	20/06 @ 17:30	13	8	outside of scope	5
Journal of International Development	https://onlinelibrary.wiley.com/journal/10991328	earthquake	20/06 @ 17:00	19	19	outside of scope	0
Journal of Refugee Studies	https://academic.oup.com/jrs	earthquake	20/06 @ 17:10	16	16	not relevant	0

Database name	URLs	Search string used (use one line per search string)	Date & time of the search	Total number of hits	Number of excluded hits - Step 1	Main reasons for exclusion after Step 1	In Zotero
Journal of Development Studies	https://www.tandfonline.com/toc/fjds20/current	earthquake	29/06 @ 17:20	44	44	natural disasters focused not earthquakes in particular	0
Journal of Humanitarian Assistance	http://journals.sagepub.com/home/eau	earthquake	17/06 @ 14:09	10	10	no lessons learnt	0
Environment and Urbanization	http://journals.sagepub.com/home/eau	earthquake and lessons	29/06 @ 17:41	1	0	outside of scope	1
Forced Migration Review	https://www.fmreview.org	earthquake	29/06 @ 17:30	10	10	outside of scope	0
ODI	http://www.odi.org/projects/	disaster response (no free entry, only list of options to choose from)	19/06 @ 16:24	41	40	not relevant	1
Governance and Social Development Resource Centre (GSDRC)	http://www.gsdr.org/publications/	earthquake	18/06 @ 14: 27	8	8	not relevant	0
Abdul Latif Jameel Poverty Action Lab (JPAL)	https://www.povertyactionlab.org/fr	earthquake and humanitarian and lessons	18/06 @ 14: 23	1	1	outside of scope	0
ALNAP HELP	https://www.alnap.org/help-library	earthquake AND humanitarian AND lessons	18/06 @ 16:30	123	107	not relevant	16
International Federation of Red Cross and Red Crescent Societies (IFRC)	www.ifrc.org	earthquake and humanitarian and lessons	18/06 @ 13:17	26	21	outside of scope	5
United Nations High Commissioner for Refugees (UNHCR)	unhcr.org	earthquake	18/06 @ 11:02	674	674	opinion pieces/stories, no research documents or evaluations	0
UN Habitat	unhabitat.org	earthquake	18/06 @13:14	0	0	no lessons learnt	0
UNICEF	unicef.org	earthquake and lessons learnt and humanitarian response	28/06@13:58	799	797	no lessons learnt	2

Database name	URLs	Search string used (use one line per search string)	Date & time of the search	Total number of hits	Number of excluded hits - Step 1	Main reasons for exclusion after Step 1	In Zotero
OCHA	www.unocha.org	No library, only theme search, did not include earthquake	N/A	0	0	N/A	0
WFP	https://www.wfp.org	No library, only theme search, did not include earthquake	N/A	0	0	N/A	0
Habitat for Humanity	https://www.interaction.org/work/shelter	earthquake	18/06 @ 9:40	112	112	stories, opinion pieces, no research documents or evaluations	0
InterAction-led shelter group	https://www.interaction.org/work/shelter	earthquake	18/06 @ 14:54	8	8	opinion pieces	0
Food Security Cluster	http://foodsecuritycluster.net	earthquake and lessons	28/06 @ 11:32	5	3	webinars/minutes from meeting	2
Education Cluster	http://educationcluster.net	earthquake	28/06@11:12	12	11	no lessons paper	1
WASH Cluster	http://washcluster.net	earthquake	28/06@11:02	0	0	no lessons paper	0
CCCM Cluster	http://www.globalccmcluster.org/tools-and-guidance/publications	earthquake	28/06 @ 11:01	10	10	no lessons papers	0
Protection Cluster	www.globalprotectioncluster.org	earthquake	27/06 @ 16:01	3	3	no lessons papers	0
Logistics Cluster	https://logcluster.org	earthquake	18/06 @ 15:02	13	12	not about earthquakes	1
Humanitarian Library	www.humanitarianlibrary.org	earthquake and humanitarian and lessons	18/06 @ 15:22	11	8	outside of scope (Pakistan)	3
Action contre la Faim	www.actioncontrelafaim.org	No library	N/A	0	0	N/A	0
Bill & Melinda Gates Foundation	www.gatesfoundation.org	No library	N/A	0	0	N/A	0
Care International	https://www.care-international.org	earthquake	18/06 @ 15:54	25	25	no lessons learnt paper	0

Database name	URLs	Search string used (use one line per search string)	Date & time of the search	Total number of hits	Number of excluded hits - Step 1	Main reasons for exclusion after Step 1	In Zotero
International Rescue Committee	https://www.rescue.org	earthquake	27/06 @ 14:41	1	1	no lessons learnt	0
EMMA toolkit	https://www.emma-toolkit.org	earthquake	18/06 @ 18:10	11	11	no lessons learnt	0
MercyCorps	https://www.mercycorps.org.uk	earthquake	27/06 @ 14:38	50	50	no lessons learnt	0
Oxfam	http://policy-practice.oxfam.org.uk/publications	earthquake	19/06 @ 15:10	17	9	no lessons learnt	8
Médecins Sans Frontières	https://www.msf.fr	earthquake	27/06 @ 14:35	5	5	no lessons learnt, only activities	0
World Vision	https://www.wvi.org/publications/129	earthquake AND lessons	27/06 @ 14:16	25	25	no lessons learnt, only activities	0
Save the Children	http://resourcecentre.savethechildren.se	earthquake	19/06 @ 15:30	67	63	no lessons learnt	4
DFID	http://r4d.dfid.gov.uk/	earthquake	18/06 @ 18:03	5	5	not relevant	0
ECHO	https://euaidexplorer.ec.europa.eu	earthquake and lessons	19/06 @ 16:40	43	40	not relevant	3
USAID	https://www.usaid.gov/data	earthquake	19/06 @ 17:15	74	72	outside of scope	2
Urgence Rehabilitation Developpement	http://www.urd.org	earthquake and lessons	29/06 @12:05	64	62	outside of scope	2
TOTAL				3971	3831		140

2.5 Exclusion table

These are the 22 studies that were excluded (and the reason why) after full screening of documents.

Table 4: Articles that were excluded and reasons for exclusion

Number	Article reference	Reason for exclusion after step 2
1	Capdevila, M., Maas, S. and Zimmermann, S. (2014) <i>Evaluation of Civil Protection Mechanism - Case study report - Earthquake Japan 2011</i> . ICF International (https://ec.europa.eu/echo/files/evaluation/2015/CPM_case_study_japan_en.pdf)	Lessons learnt are too specific to the EU, not to humanitarian practitioners
2	Chung, E. (2017) 'The outcomes and impact of a post-earthquake rehabilitation program in China: a qualitative study'. <i>Qualitative Health Research</i> , 27(2): 170-181	No lessons learnt
3	Fitter, D.L., Delson, D.B., Guillaume, F.D., Schaad, A.W., Moffett, D.B., Poncet, J.L., Lowrance, D. and Gelting, R. (2017) 'Applying a new Framework for public health systems recovery following emergencies and disasters: the example of Haiti following a major earthquake and cholera outbreak'. <i>The American Journal of Tropical Medicine and Hygiene</i> , 97(S4): 4-11	Outside of scope, contains lessons on how to respond to a cholera epidemic, not how to respond to an earthquake
4	Cohen, S. (2011), 'What actions and policies can the government of Haiti implement to improve Emergency Management Response	Outside of scope
5	Irwin, B. and Horugavye, P. (2012) <i>Three years on - Haiti earthquake response</i> . World Vision (https://www.wvi.org/haiti/publication/three-years-haiti-earthquake-response-report)	No lessons learnt
6	Katwal, N. (2015) 'Post disaster livelihood recovery: a case study of the 2015 earthquake in Nepal'. Master thesis (https://brage.bibsys.no/xmlui/handle/11250/2401013)	No lessons learnt
7	Katsube, T. (2012) <i>The Great Japan Earthquake and Tsunami - response and lessons learned</i> . Japan International Cooperation Agency (https://www.alnap.org/system/files/content/resource/files/main/day1-tkatsube.pdf)	Power point presentation
8	Shelter Cluster Nepal (2015) <i>Shelter Cluster Nepal position paper, housing and settlements recovery: the need for a support approach</i> . Shelter Cluster Nepal (https://www.sheltercluster.org/sites/default/files/docs/nepal_recovery_2015_07092015.pdf)	No lessons learnt

Number	Article reference	Reason for exclusion after step 2
9	Grünewald, F. (2013) 'Leçons tirées d'anciennes catastrophes naturelles qui peuvent être utiles pour la réponse au tremblement de terre en Haïti' (https://www.urd.org/Lecons-tirees-d-anciennes?artpage=2-2#outil_sommaire_9)	Blog article
10	ALNAP (2010) <i>Final Independent Evaluation of the 'Emergency Intervention in Nutrition and Mental Health - Earthquake in Port-au-Prince (Haïti)' programme</i> . London: ALNAP/ODI (https://www.alnap.org/help-library/final-independent-evaluation-of-the-emergency-intervention-in-nutrition-and-mental)	ToRs only
11	International Organization for Migration (IOM) Haiti (2012) <i>Haiti: from emergency to sustainable recovery. IOM Haiti two year report (2010-2011)</i> (https://reliefweb.int/sites/reliefweb.int/files/resources/Full%20Report_645.pdf)	No lessons learnt
12	Shenhar, G., Adamcheck, R. and Hopmeier, M. (2015) 'The need for international search and rescue (SAR) teams during an earthquake: Nepal case study'. <i>Development in Practice</i> , 26(7): 949-953	No lessons learnt
13	Abrahams, D. (2014) 'The barriers to environmental sustainability in post-disaster settings: a case study of transitional shelter implementation in Haiti'. <i>Disasters</i> , 38(1): 25-49	No lessons learnt
14	Yang, F., Yuan, Q., Du, S. and Liang, L. (2014) 'Reserving relief supplies for earthquake: a multi-attribute decision making of China Red Cross', <i>Annals of Operations Research</i> , 247(2): 759-785	Outside of scope + no lessons learnt
15	Shelter Cluster Nepal Recovery and Reconstruction Working Group (2015) 'Experience and lessons sharing from Pakistan and Aceh 8th October 2015'. (https://www.sheltercluster.org/sites/default/files/docs/lessons_learned_and_experience_sharing_pakistanaceh_8th_october_2015.pdf)	Minutes from Meeting
16	Pascapurnama, D.N., Murakami, A., Chagan-Yasutan, H., Hattori, T., Sasaki, H. and Egawa, S. (2017) 'Integrated health education in disaster risk reduction: lesson learned from disease outbreak following natural disasters in Indonesia'. <i>International Journal of Disaster Risk Reduction</i> , 29: 94-102	Outside of scope, preparedness only
17	Pyles, L., Svistova, J. and André, J. (2015). <i>Disaster recovery in post-earthquake rural Haiti: research findings and recommendations for participatory, sustainable recovery</i> . School of Social Welfare, State University of New York at Albany (https://www.albany.edu/ssw/assets/HaitirecoveryreportNSF_final.pdf)	No lessons learnt

Number	Article reference	Reason for exclusion after step 2
18	Mattson, K. (2013) <i>Technical review of water, sanitation and hygiene promotion activities for t-shelter beneficiaries</i> . IFRC (https://www.alnap.org/help-library/technical-review-of-water-sanitation-and-hygiene-promotion-activities-for-t-shelter)	Outside of scope + no lessons learnt
19	Levie, F., Burke, C.M. and Lannon, J. (2017). 'Filling the gaps: an investigation of project governance in a non-governmental organisation's response to the Haiti earthquake disaster'. <i>International Journal of Project Management</i> , 35(5): 875-888	Outside of scope
20	Chen, Y. and Booth, D.C. (2011) <i>The Wenchuan earthquake of 2008: anatomy of a disaster</i> . Beijing and Berlin: Science Press and Springer-Verlag	Outside of scope
21	Yanev, P. (2010) "It is not too late": preparing for Asia's next big earthquake with emphasis on the Philippines, Indonesia, and China: policy note. World Bank and Global Facility for Disaster Reduction and Recovery (GFDRR) (http://documents.worldbank.org/curated/en/352601468026941451/pdf/576830WP0REVIS1AVOIDIEARTHQUAKE1web.pdf)	Outside of scope
22	Wilder, A. (2010) 'Aid and stability in Pakistan: lessons from the 2005 earthquake response'. <i>Disasters</i> , 34: s406-s426	Outside of scope

2.6 Data extraction matrix

Once irrelevant studies had been excluded, the reviewers coded studies that met the criteria for inclusion using the coding sheet below. Data was entered manually into Microsoft Excel in order to facilitate comparison and analysis. Reviewers inserted short quotes in order to illustrate important points.

Table 5: Coding of studies

Code	Answer	Comments
Admin codes		
Citation of the study		
Region		
Domain of the study	Academia Grey literature	
Type of study	Journal article Research report Evaluation report Conference paper Book/chapter Thesis/dissertation M&E report	
Research question		
Linked studies		
Context codes		
Where was the response conducted?	LICs LMIC UMICs State country:	
What has been the impact of the crisis on physical infrastructures?		
How long after the earthquake did the response start?	Less than a month Between 1 and 3 months Between 3 and 6 months	
How many sites? Response sites? Experimental sites?		
What was the setting?	Rural Urban Mixed Camp	

Population codes		
How many people were targeted by the earthquake response activity?		
Age	Children (<12) Youth (12-25) Adults (25-60) Elderly (>60) Mixed	
Gender	Male Female Both	
Literacy	Literate Semi-literate Illiterate	
Specific health-related status	Living with disability Pregnant women Chronic diseases Other	
Status	Refugee Internally displaced person Host communities Local communities	
How many crisis-affected people were impacted indirectly by the earthquake response activities?		
Age	Children (<12) Youth (12-25) Adults (25-60) Elderly (>60) Mixed	
Gender	Male Female Both	
Specific health-related status	Living with disability Pregnant woman Chronic diseases Other	
Status	Refugee Internally displaced person Host communities Local communities	
Level of income		
Response codes		
What was the sector of the response?	Debris Food security WASH education nutrition health livelihood shelter protection Intervention aiming at covering multiple needs	
Who initiated the response?	NGOs UN agencies Red Cross/ Red Crescent movement Private sector actors Government actors	
What activities were implemented?		
What modality was used?	CTP support In-kind support Services (transport, storage or else) Skills development Other	

Outcome codes			
Which outcomes were targeted as regard to the crisis-affected population?		– Improved shelter condition – Increased food security – Restored livelihoods – Improved access to water, sanitation or hygiene – Infrastructure reconstruction – Road clearing – Improved nutritional status – Improved protection – Increased access to education for children – Reduced number of negative coping mechanisms – Other outcomes	
What outcome indicators were used?	And how were they measured?	– Sector-specific indicator (e.g. food consumption score) – Proxy indicator – Coping strategy index – Other	– Household survey – Key informant interview – Focus group discussion – Desk review – Other
When were the outcomes measured?		Pre-test: Post-test:	

Findings		
What findings do they report?		
Post-earthquake response effectively improved the situation of crisis-affected population		
Post-earthquake response had no impact on the situation of crisis-affected population		
Post-earthquake response had a negative impact on crisis-affected population (describe)		
Post-earthquake response had an impact on the situation of the local markets		
Post-earthquake response had an impact on the situation of non-targeted households		
Post-earthquake response had an impact on the overall humanitarian response in country		
Were the activities feasible?	Yes ('but' if applicable) No ('because') Lack of access Lack of community acceptance Lack of political acceptance Donor reluctance	
How have the post-earthquake activities been received?	By the affected population: Positive Negative Mixed No information	
Describe the situation after the response		
Key lessons across the project cycle		
Needs assessment		
Strategic planning		
Resource mobilisation		
Implementation and monitoring		
Operational peer review and evaluation		
Coordination		
NOTE		

2.7 Critical appraisal tool

Study type: qualitative Methodological appraisal criteria	Score	Comment / Confidence judgement
1. Study is defensible in design Appraisal indicators: Is the study design clearly specified and appropriate for aims and objectives of the research?		
Consider whether:		
– there is a discussion of the rationale for the study design		
– the research question is clear, and suited to qualitative inquiry		
– there are convincing arguments for different features of the study design		
– limitations of the research design and implications for the research evidence are discussed		
Total score	Worth to continue?	
2. Study features an appropriate sample Appraisal indicators		
Consider whether:		
– there is a description of study location and how/why it was chosen		
– the author has explained how the participants were selected		
– the selected participants were appropriate to collect rich and relevant data		
– reasons are given why potential participants chose not to take part in study		
Total score	Worth to continue?	

3. Study is rigorous in conduct		
Appraisal indicators		
Consider whether:		
– the author provides a clear account/description of the process by which data was collected (e.g. for interview method, is there an indication of how interviews were conducted?/ procedures for collection or recording of data?)		
– the author demonstrates that data collection targeted depth, detail and richness of information (e.g. interview/observation schedule)		
– there is evidence of how descriptive analytical categories, classes, labels, etc. have been generated and used		
– presentation of data distinguishes clearly between the data, the analytical frame used, and the interpretation		
– if methods were modified during the study, has the author explained how and why?		
Total score	Worth to continue?	

4. Study findings are credible in claim/based on data Appraisal indicators		
Consider whether:		
– there is a clear description of the form of the original data		
– sufficient amount of data is presented to support interpretations and findings/conclusions		
– the author explains how the data presented were selected from the original sample to feed into the analysis process		
– there is a clear and transparent link between data, interpretation, and findings/conclusion		
– there is evidence (of attempts) to give attention to negative cases/outliers etc.		

Total score

If findings not credible, can data still be used?

5. Study attends to contexts Appraisal indicators		
Consider whether:		
– there is an adequate description of the contexts of data sources and how they are retained and portrayed		
– participants' perspectives/observations are placed in personal contexts		
– appropriate consideration is given to how findings relate to the contexts (how findings are influenced by or influence the context)		
– the study makes any claims (implicit or explicit) that infer generalisation (if yes, comment on appropriateness)		

Total score

6. Study is reflexive Appraisal indicators			
Consider whether:			
– appropriate consideration is given to how findings relate to author's influence/own role during analysis and selection of data for presentation			
– the author has attempted to validate the credibility of findings (e.g. triangulation, respondent validation, more than one analyst)			

Total score

NB: Can override previous exclusion!

OVERALL DECISION – EXCLUDE / INCLUDE

(study generates new knowledge relevant to the review question and complies with minimum criteria to ensure reliability and empirical grounding of knowledge)

Study type: quantitative Methodological appraisal criteria		Comment / risk of bias judgement	Score
1. Selection bias Appraisal indicators			
Consider whether:			
– there is a clear description of how and why sample was chosen			
– there is adequate sample size to allow for representative and/or statistically significant conclusions			
– participants recruited in the control group were sampled from the same population as that of the treatment group			
– group allocation process attempted to control for potential risk of bias			

Total score

Worth to continue?

2. Bias due to missing data Appraisal indicators			
Consider whether:			
– outcome data are reasonably complete (80% or above)			
– If 'no', are missing data reported?			
– If missing data: are proportion of participants and reasons for missing data similar across groups?			
– If missing data: were appropriate statistical methods used to account for missing data? (e.g. sensitivity analysis)			
– If not possible to control for missing data, are outcomes with missing data excluded from analysis?			

Total score

Worth to continue?

3. Outcome reporting bias Appraisal indicators		
Consider whether:		
– the outcome measure was clearly defined and objective		
– outcomes were assessed using standardised instruments and indicators		
– outcome measurements reflect what the experiment set out to measure		
– the methods of outcome assessment were comparable across experiential groups		
Total score	Worth to continue?	

OVERALL RISK OF BIAS		
Study type: Mixed-method Methodological appraisal criteria	Score	Comment /confidence judgement
Research integration/synthesis of methods Appraisal indicators		
Consider whether:		
– The rationale for integrating qualitative and quantitative methods to answer the research question is explained		
– The mixed-methods research design is relevant to address the qualitative and quantitative research questions, or the qualitative and quantitative aspects of the mixed-methods research question		
– There is evidence that data gathered by both research methods was brought together to inform new findings to answer the mixed-methods research question (e.g. form a complete picture, synthesise findings, configuration)		
– The approach to data integration is transparent and rigorous in considering all findings from both the qualitative and quantitative module (danger of cherry-picking)		
– Appropriate consideration is given to the limitations associated with this integration, e.g. the divergence of qualitative and quantitative data (or results)		
For mixed-methods research studies, each component undergoes its individual critical appraisal first. Since qualitative studies are either included or excluded, no combined risk of bias assessment is facilitated, and the assigned risk of bias from the quantitative component similarly holds for the mixed-methods research. The above appraisal indicators only refer to the applied mixed-methods design. If this design is not found to comply with each of the four mixed methods appraisal criteria below, then the quantitative/qualitative components will individually be included in the review.		
<u>Mixed-methods critical appraisal:</u> 1. Study is defensible in design 2. Study is rigorous in conduct 3. Study is credible in claim 4. Study is reflective	<u>Qualitative critical appraisal:</u> – Include / Exclude	<u>Quantitative critical appraisal:</u> 1. Low risk of bias 2. Risk of bias 3. High risk of bias 4. Critical risk of bias
Combined appraisal: Include / Exclude mixed methods findings judged with _____ risk of bias		

* Scoring Matrix: 0 = No; 1 = Partly; 2 = Yes

2.8 Data synthesis table

This table details which papers are cited in which lesson, the confidence scoring of each lesson and the confidence scoring matrix.

To obtain the confidence scoring, we took the following steps. First, we averaged the quality score of each document in each lesson to obtain a raw quality score (out of 48) and then counted the number of documents contributed to each lesson to capture the size of the evidence base. Using these scores, we then calculated the means for both the quality and the evidence base. The average quality score for a lesson is 23 out of 48 and the average number of papers cited in each lesson is 7. Based on these results, we then calculated the deviation of each lesson score to the mean.

Using a ranking matrix capturing both the deviation in terms of quality and quantity, we attributed a four-level qualitative score (low confidence, medium confidence, high confidence, extremely high confidence), which we then translated into a quantitative score (low confidence=1, medium confidence=2, high confidence=3, very high confidence=4)

Key for data synthesis table

Quality rating		Evidence base	
Deviation to the mean	Score	Deviation to the mean	Score
-12 to -7	Low quality rating	-5 to -2	Low evidence base
-6 to -3	Medium quality rating	-2 to +2	Medium evidence base
-2 to +1	High quality rating	+2 to +4	High evidence base
+2 to +8	Very high quality rating	+4 to +6	Very high evidence base

		Low evidence base	Medium evidence base	High evidence base	Very high evidence base
		Evidence base			
Low quality rating	Quality rating	Low confidence	Medium confidence	Medium confidence	High confidence
Medium quality rating		Medium confidence	Medium confidence	High confidence	High confidence
High quality rating		Medium confidence	High confidence	High confidence	Very high confidence
Very high quality rating		High confidence	High confidence	Very high confidence	Very high confidence

Source/Lesson	1. Engage broadly and rapidly with local and national actors – even the most affected communities and authorities have some level of capacity after an earthquake.	2. Acknowledge and do not undermine the work of private sector entities to ensure business continuity following an earthquake.	3. Do not let infrastructure and access challenges get in the way of communication, especially with isolated communities.	4. Ensure that cross-cutting issues such as gender, security and the environment are incorporated at all stages of the response.	5. Conduct thorough assessments which recognise and identify the distinct ways earthquakes affect different populations.	6. Follow established good practice for needs assessments, recognising the specific challenges of doing so in urban areas.	7. Conduct assessments in ways that avoid exacerbating tensions between host communities and internally displaced persons.	8. Locate spaces to store debris and, if appropriate, use short-term conditional assistance to clear it.	9. Anticipate issues related to lack of documentation and complex land tenure.	10. Mobilise sufficient and appropriate surge capacities.	11. Account for quickly evolving health needs in post-earthquake contexts. Be aware that epidemics can happen.	12. Prioritise the repair of existing structures, support owner-driven reconstruction, preserve architectural heritage and use relocation and resettlement only as a last resort.	13. Be cautious with setting up transitional shelters, which may hinder longer-term reconstruction	14. Address long-term as well as immediate education needs.	15. Recognise the value of MEAL and push through obstacles.	16. Put time and effort into coordination to avoid negative impacts on the response.
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Total Quality score	202	163	237	158	129	103	32	217	102	66	177	203	192	149	167	346
Evidence base	10	6	9	9	5	4	2	12	4	2	7	8	8	6	8	14
Final Score out of 48	20	27	26	18	26	26	16	18	26	33	25	25	24	25	21	25
Quality rating mean	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24
Evidence base mean	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
Deviation to the mean (quality rating)	-4	3	3	-6	2	2	-8	-6	2	9	2	2	0	1	-3	1
Qualitative score - Quality rating	Medium quality rating	Very high quality rating	Very high quality rating	Medium quality rating	Very high quality rating	High quality rating	Low quality rating	Medium quality rating	Very high quality rating	High quality rating	High quality rating	Very high quality rating	High quality rating	High quality rating	High quality rating	Very high quality rating
Deviation to the mean (evidence base)	3	-1	2	2	-2	-3	-5	5	-3	-5	0	1	1	-1	1	7
Qualitative score - Evidence base	High evidence base	Medium evidence base	Medium evidence base	Medium evidence base	Medium evidence base	Low evidence base	Low evidence base	Very high evidence base	Low evidence base	Low evidence base	Medium evidence base	Medium evidence base	Medium evidence base	Medium evidence base	Medium evidence base	Very high evidence base
Qualitative confidence score	High confidence	High confidence	High confidence	Medium confidence	High confidence	Medium confidence	Low confidence	High confidence	High confidence	Medium confidence	High confidence	High confidence	High confidence	High confidence	High confidence	Very high confidence
Quantitative confidence score (out of 4)	3	3	3	2	3	2	1	3	3	2	3	3	3	3	3	4



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